PaN-Finder

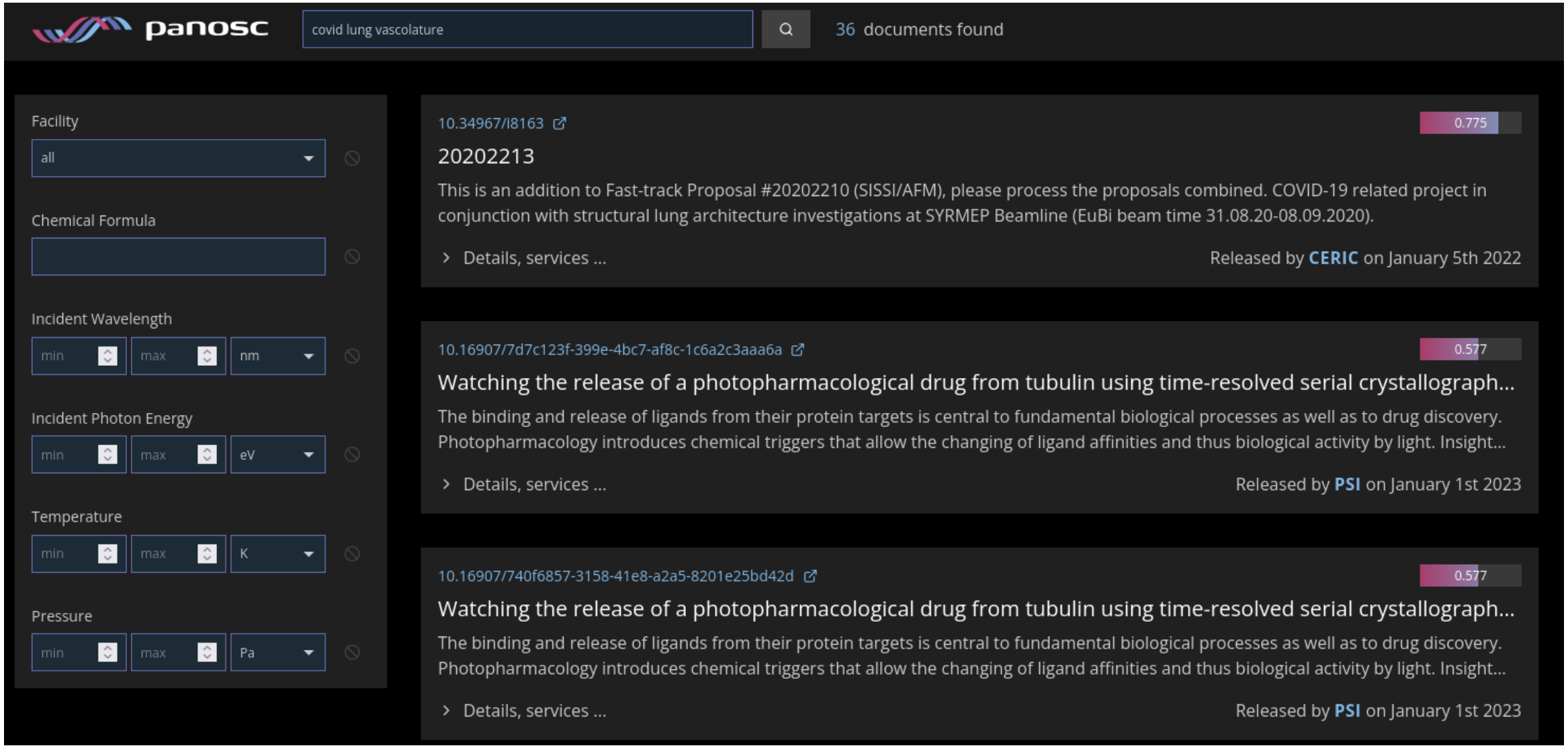
Max Novelli and the SIMS team

DMSC Division Meeting , 2025-08-18

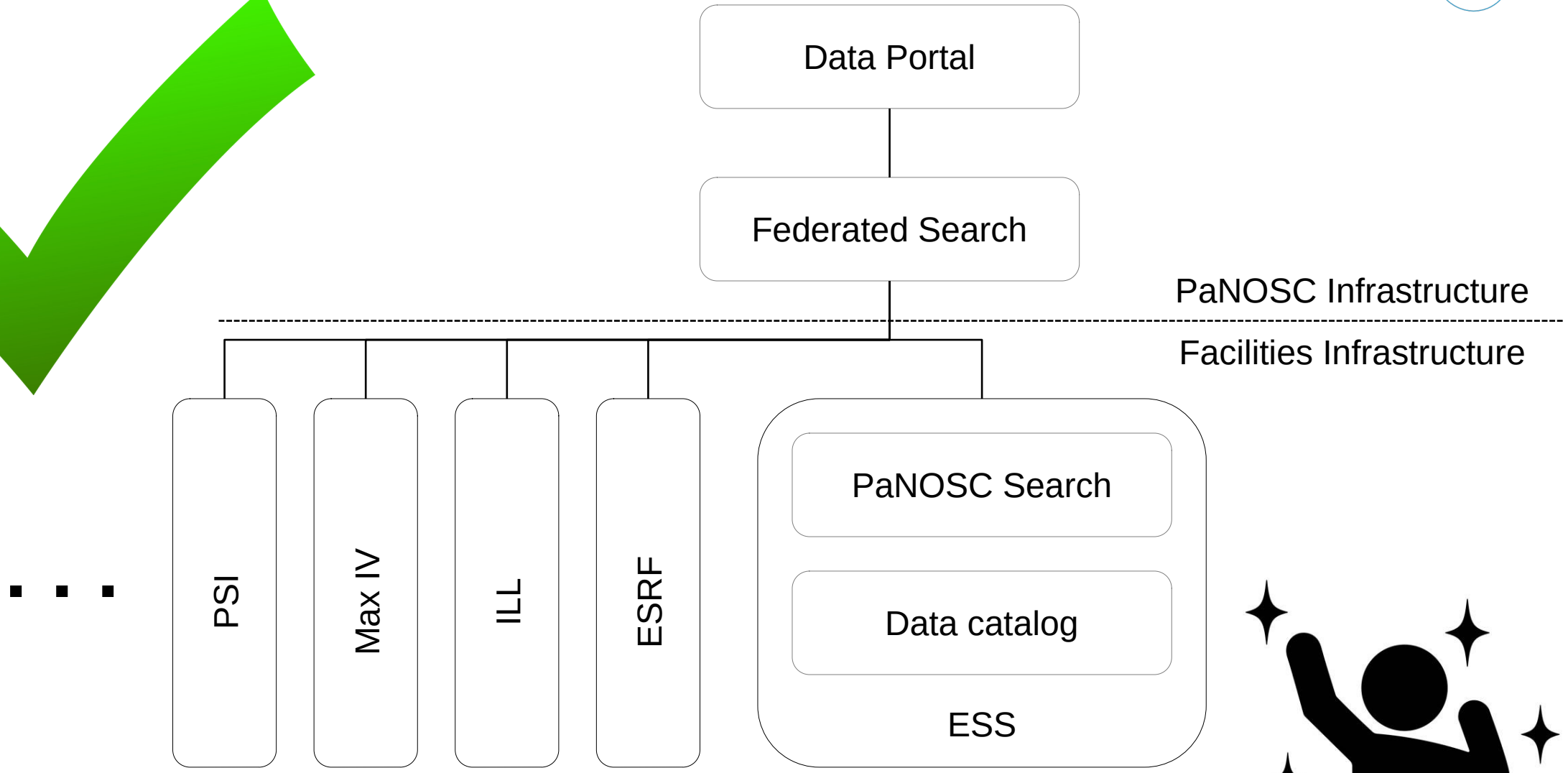
The beginning

Greg Razoky on Unsplash

PaNOSC Data Portal



Success!!!


Data Portal
Federated Search
PaNOSC Infrastructure
Facilities Infrastructure
. . .
PSI
Max IV
ILL
ESRF
PaNOSC Search
Data catalog
ESS

Few Challenges

- The project ended.
  Supported as in-kind contribution

- Connection between federated and local search APIs is not robust

- Local search APIs are hard to monitor

- Data curation at each facility varies a lot

User Experience

panosc

covid lung vascolature

36 documents found

Facility

all

Chemical Formula

Incident Wavelength

min max nm

Incident Photon Energy

min max eV

Temperature

min max K

Pressure

min max Pa

10.34967/i8163 0.775

20202213

This is an addition to Fast-track Proposal #20202210 (SISSI/AFM), please process the proposals combined. COVID-19 related project in conjunction with structural lung architecture investigations at SYRMEP Beamline (EuBi beam time 31.08.20-08.09.2020).

Details, services ... Released by **CERIC** on January 5th 2022

10.16907/7d7c123f-399e-4bc7-af8c-1c6a2c3aaa6a 0.577

Watching the release of a photopharmacological drug from tubulin using time-resolved serial crystallograph...

The binding and release of ligands from their protein targets is central to fundamental biological processes as well as to drug discovery. Photopharmacology introduces chemical triggers that allow the changing of ligand affinities and thus biological activity by light. Insight...

Details, services ... Released by **PSI** on January 1st 2023

10.16907/740f6857-3158-41e8-a2a5-8201e25bd42d 0.577

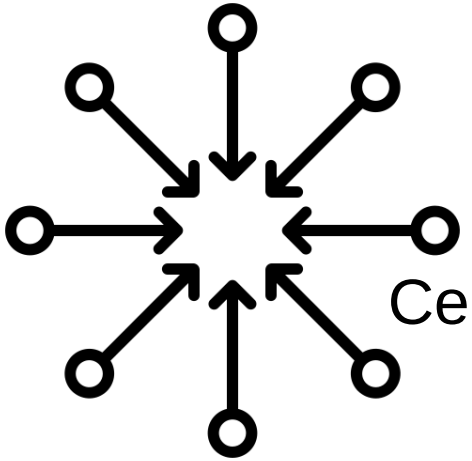
Watching the release of a photopharmacological drug from tubulin using time-resolved serial crystallograph...

The binding and release of ligands from their protein targets is central to fundamental biological processes as well as to drug discovery. Photopharmacology introduces chemical triggers that allow the changing of ligand affinities and thus biological activity by light. Insight...

Details, services ... Released by **PSI** on January 1st 2023

The Vision

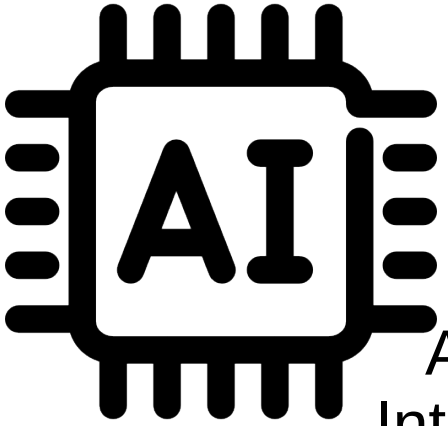
The requirements



Centralized

Asynchronous data collection

Natural Language



Artificial Intelligence

Retrieval Augmented Generation

The idea

I'm writing a new proposal for a new experiment to study the changes in molecular structure of a new synthetic carbon based material bonding with human blood red cells. I'm interested in running the experiments at a normal human body temperature. Can you show me related publications and similar proposals? Have similar experiments already been conducted?

In-vivo experiments with synthetic carbon based materials has been already performed and you can find information about them in the following resources:

- Resource 1 (URL 1)
- Resource 2 (URL 2)

I have not found information regarding new synthetic carbon based materials bonding with human blood red cells, but relevant information might be found in the following resources:

- Resource 3 (URL 3)
- Resource 4 (URL 4)

You also might want to check the following open datasets as they have been acquired from human blood red cells at normal body temperature:

- Dataset 1, RI, DOI
- Dataset 2, RI, DOI
- Dataset 3, RI, DOI

The opportunity

OSCARS

Open Science Clusters' Action for Research & Society

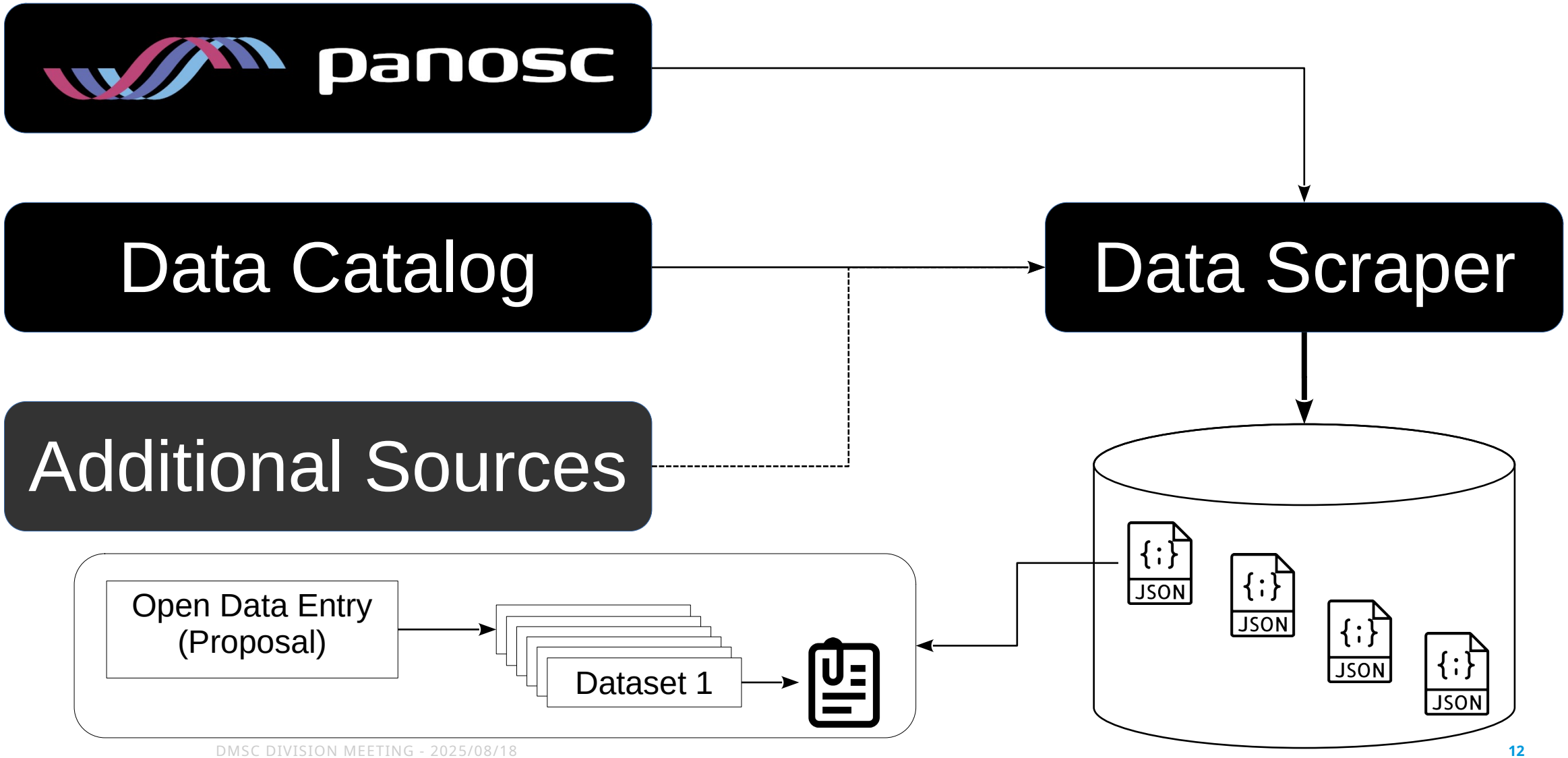
https://oscars-project.eu/

https://oscars-project.eu/projects/pan-finder-photon-and-neutron-federated-knowledge-finder

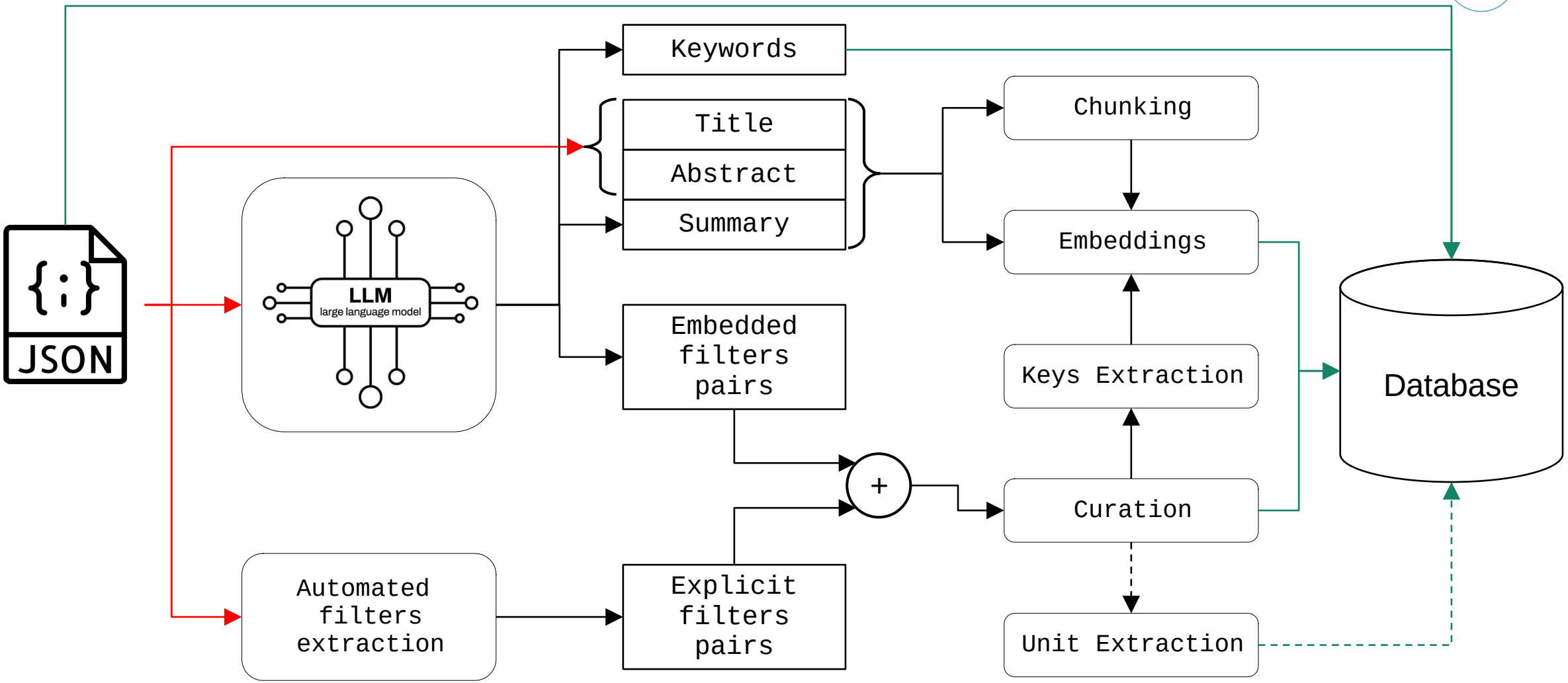
Implementation

https://lifebalancestudio.co.uk/the-reality-of-yoga-poses/

Data Collection

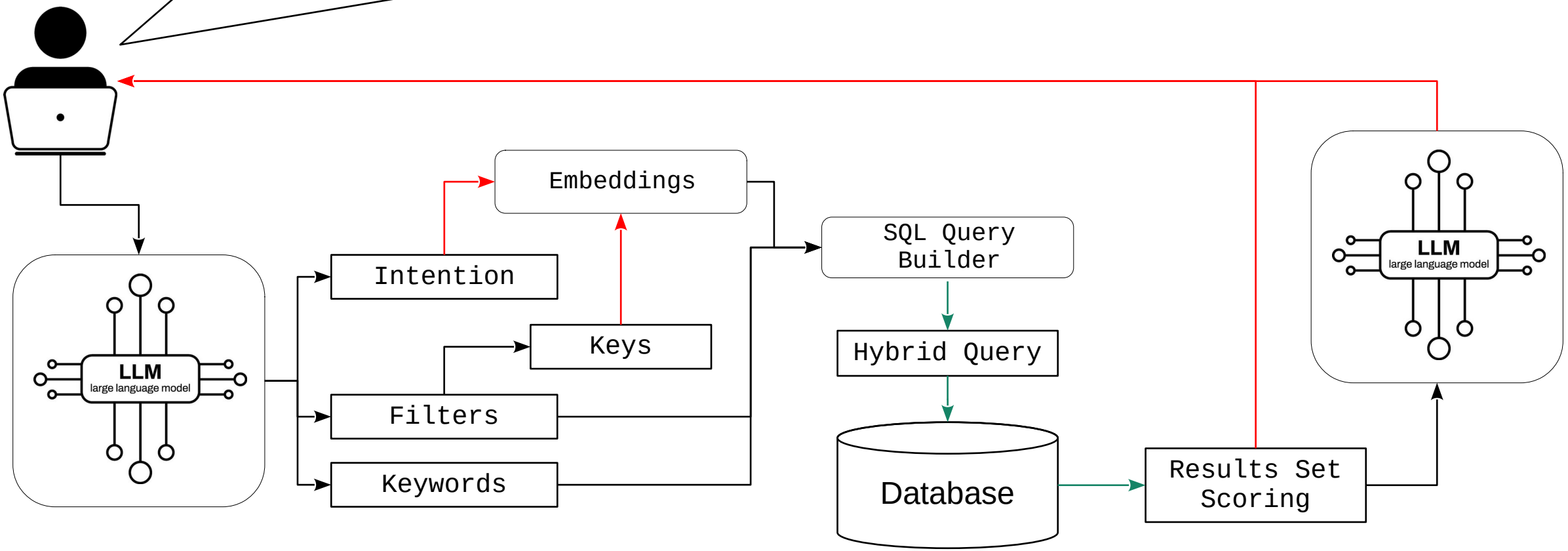


Data Ingestion



User journey

I'm writing a new proposal for a new experiment to study the changes in molecular structure of a new synthetic carbon based material bonding with human blood red cells. I'm interested in running the experiments at a normal human body temperature. Can you show me related publications and similar proposals? Have similar experiments already been conducted?



Beta version

https://pan-finder.

PaN-Finder BETA

Search through scientific datasets and research publications

I'm writing a new proposal for a new experiment to study the changes in molecular structure of a new synthetic carbon based material bonding with human blood red cells. I'm interested in running the experiments at a normal human body temperature. Can you show me related publications and similar proposals? Have similar experiments already been conducted?

New Search

Relevance Explanation

Relevant Results

- The article *Effect of oxidative stress on polyunsaturated lipid bilayers modulated by cholesterol and head group composition* **fully aligns with** your interest in understanding structural changes in cell membranes under specific conditions, as it explores how oxidative stress and nanoparticles affect the stability and structure of mammalian cell membrane models—closely paralleling your focus on molecular interactions with blood cells at physiological temperature. (DOI: 10.5291/ILL-DATA.9-13-806)
- The study *Studying the interactions of nanoparticles with tethered lipid bilayers.* **fully aligns with** your proposal by examining how engineered nanoparticles interact with model cell membranes, using advanced techniques to analyze molecular-level interactions relevant to synthetic materials bonding with biological membranes. (DOI: 10.5291/ILL-DATA.9-13-732)
- The publication *Structural investigation of Hyaluronan / Pluronic aggregates: a possible system for drug delivery with specific targeting* **fully aligns with** your interest in molecular structure and targeted interactions, as it investigates how synthetic aggregates interact with cell membranes for drug delivery, offering insights into the structural characterization of such systems in a biomedical context. (DOI: 10.5291/ILL-DATA.9-13-564)

Additional Background & Context

- The research *Understanding the interfacial behaviour of bile salts to better engineer lipid emulsions: neutron reflectivity measurements* provides useful background by exploring how biomolecules interact with lipid films, shedding light on interfacial structural changes that may inform your understanding of synthetic material interactions with biological membranes. (DOI: 10.5291/ILL-DATA.9-13-741)
- The study *Coated therapeutic target nanoparticles - D-contrast SANS study of mixed coating stabilization* offers additional context by detailing the design and structural analysis of nanoparticles for targeted therapy, which, while focused on cancer treatment, discusses techniques and concepts relevant to the stability and interaction of synthetic materials with biological systems. (DOI: 10.5291/ILL-DATA.9-12-428)

Most Relevant Documents

Click rows to view details

20 found

DOI	Title	Facility
10.5291/ILL-DATA.9-13-806	Effect of oxidative stress on polyunsaturated lipid bilayers modulated by cholesterol and head group composition	ILL
10.5291/ILL-DATA.9-13-732	Studying the interactions of nanoparticles with tethered lipid bilayers.	ILL
10.5291/ILL-DATA.9-13-564	Structural investigation of Hyaluronan / Pluronic aggregates: a possible system for drug delivery with specific targeting	ILL
10.5291/ILL-DATA.9-10-1485	Squalene Nanomedecine: solvent effect in controlling size and internal structure	ILL
10.5291/ILL-DATA.9-13-741	Understanding the interfacial [illegible] better engineer lipid emulsions: neutron reflectivity measurements	ILL

Show all 20 results

User Prompt Components

I'm writing a new proposal for a new experiment to study the changes in molecular structure of a new synthetic carbon based material bonding with human blood red cells. I'm interested in running the experiments at a normal human body temperature. Can you show me related publications and similar proposals? Have similar experiments already been conducted?

New Search

```
{
  "intention": "publications and proposals on molecular structure of synthetic carbon based material bonding with human blood red cells",
  "keywords": [
    "molecular structure",
    "synthetic carbon based material",
    "human blood red cells"
  ],
  "filters": {
    "logic": "AND",
    "conditions": [
      {
        "name": "temperature",
        "operator": "=",
        "value": 37,
        "unit": "°C"
      }
    ]
  }
}
```

Demo

What’s next?

Current and future work

- Benchmarking
- Testing different models (size, cost, performance)
- Prompt Engineering
- Model fine tuning
- Refine Natural Language Response
- Automating Data Collection
- Driving better data curation

Thank you for your attention

Thanks to Janos Babik
and the SIMS team

Questions?

Max Novelli
max.novelli@ess.eu